

Notes: Spiegel (RES, 2026)

Behavioural Causal Inference

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1 Big picture

- **Question.** How costly are bad controls when decision-makers learn causal effects from endogenous observational data and then act on the correlations they see?
- **Motivation.** Professionals and lay people often infer causal effects by controlling for variables. But controlling for the wrong variables can create biased causal inference. The paper asks how large the resulting welfare cost can be in equilibrium.
- **Key behavioural feature.** The decision-maker behaves like a frequentist user of a historical database. He observes correlations between actions, outcomes, and some controls, then treats the controlled correlation as causal.
- **Key equilibrium feature.** The database is endogenous. Past decision-makers' actions generate the correlations that future decision-makers learn from. Hence beliefs and choices must be jointly stable.
- **Model environment.** A decision-maker chooses a binary action $a \in \{0, 1\}$, has a binary preference type $t \in \{0, 1\}$, observes some exogenous variables x , and cares about outcome y plus a cost of choosing an action that differs from t .
- **Main distinction.** The welfare cost depends sharply on whether data types' control sets are *vertically ordered* by set inclusion. Nested control sets create strong equilibrium discipline; horizontally differentiated controls can create mutually reinforcing confounding.
- **Homogeneous-preference result.** If $t = 0$ for everyone and control sets are vertically ordered, the unique equilibrium has no mistaken action and zero welfare loss. If control sets are not vertically ordered, the welfare loss can approach the non-equilibrium worst case.
- **Heterogeneous-preference result.** When t varies and $y \perp x \mid t$, vertically ordered controls imply a tight upper bound of $\gamma(1 - \gamma)$ on welfare loss. Without vertical ordering, the bound is much larger: $\max\{\gamma, 1 - \gamma\}$ under the restricted domain, and full error is possible without that restriction.
- **Adjustment extension.** The paper distinguishes conditioning on a realized variable from adjusting for a variable using long-run statistics. The same vertical-ordering logic survives after replacing set inclusion with a relation iPj defined by $D_i \supseteq C_j$.
- **Consequential-action extension.** If the action has a true separable causal effect β , the same worst-case bounds apply after multiplication by $1 - \beta$.
- **Economic takeaway.** Equilibrium can protect people from bad causal inference because mistaken actions change the data in a way that weakens the misleading correlation. But this protection is much stronger when agents' controls are nested rather than horizontally fragmented.

2 Data, variables, and measurement in the model

2.1 Agents, choices, and payoffs

The decision-maker chooses a binary action

$$a \in A = \{0, 1\}.$$

The preference type is

$$t \in \{0, 1\},$$

where t is the action the decision-maker intrinsically prefers. The outcome is

$$y \in Y \subseteq [0, 1].$$

The vector of pre-treatment exogenous variables is

$$x = (x_1, \dots, x_K).$$

The baseline utility is

$$u(t, a, y) = y - \theta \cdot \mathbf{1}\{a \neq t\}, \quad \theta \in (0, 1).$$

Interpretation. The decision-maker chooses the action that matches his taste unless he believes the other action improves the outcome enough to compensate for the taste cost θ .

2.2 The true causal structure

In the baseline model, a has no causal effect on y . Formally,

$$y \perp a \mid (t, x).$$

The joint distribution factorizes as

$$p(t, x, a, y) = p(t, x)p(a \mid t, x)p(y \mid t, x).$$

The first and third terms are exogenous; the action rule $p(a \mid t, x)$ is endogenous because it is generated by the population's equilibrium strategies.

Interpretation. Bad causal inference is possible because a and y may be correlated even when the true causal effect is zero. The correlation comes from omitted or incorrectly handled causes t, x .

2.3 Data types and controls

Each data type $i \in N = \{1, \dots, n\}$ controls for a subset

$$C_i \subseteq \{1, \dots, K\}.$$

The population share of type i is λ_i . A strategy for preference type t and data type i is

$$\sigma_{t,i} : X_{C_i} \rightarrow \Delta(A).$$

Aggregate behaviour in the public dataset is

$$p(a \mid t, x) = \sum_{i \in N} \lambda_i \sigma_{t,i}(a \mid x_{C_i}).$$

Interpretation. The public data represent the aggregate choices of all data types. A type with limited controls does not see the whole causal system; he sees only the conditional correlations available through his chosen variables.

2.4 Perceived causal effect

Type i treats C_i as a valid set of controls and estimates the causal effect of switching from $a = 0$ to $a = 1$ as

$$\Delta_i(x) = E_p(y \mid a = 1, x_{C_i}) - E_p(y \mid a = 0, x_{C_i}).$$

Interpretation. The key object is not the true causal effect, which is zero in the baseline. It is the perceived effect extracted from endogenous observational data after controlling for C_i .

2.5 Welfare loss

A mistaken action is $a \neq t$. Given a strategy profile σ , expected welfare loss is

$$L(\sigma) = \theta \sum_{t,x} p(t, x) \sum_{i \in N} \lambda_i \sigma_{t,i}(a \neq t \mid x).$$

Interpretation. Since a has no objective causal effect on y in the baseline, the only welfare loss comes from taking the action that does not match t .

3 Models and theoretical specifications

3.1 Equilibrium

An ε -equilibrium requires every action used with probability above ε to be optimal given the perceived causal effect induced by the aggregate database. In compact form,

$$a' \in \arg \max_a \{E_p(y \mid a, x_{C_i}) - \theta \mathbf{1}\{a \neq t\}\}.$$

An equilibrium is the limit of ε -equilibria as $\varepsilon \rightarrow 0$.

Interpretation. This is a steady state of behavioural causal inference. Past choices generate the dataset; the dataset generates perceived causal effects; perceived causal effects generate current choices.

3.2 Why equilibrium matters

Without the equilibrium restriction, welfare loss can be arbitrarily close to 1. For example, if $t = 0$ always, $y = x$, and decision-makers mechanically choose $a = x$, mistakes occur whenever $x = 1$. But this pattern is not an equilibrium if the type who conditions on x then realizes that a has no causal effect.

Interpretation. The endogenous dataset can undermine the very correlation that initially fooled the decision-maker. This is the central force that produces equilibrium protection.

3.3 Vertical ordering of controls

In the baseline conditioning model, data types are vertically ordered if they can be labeled such that

$$C_1 \supset \cdots \supset C_n.$$

Lower-indexed types control for larger sets of variables.

Interpretation. Vertical ordering represents environments in which all agents broadly agree on the relevant controls but differ in data access. Some have premium access to more controls; others have coarser data.

3.4 Homogeneous preferences: $t = 0$

When $\gamma = p(t = 1) = 0$, the correct action is always $a = 0$ and welfare loss is

$$L = \theta \Pr(a = 1).$$

The central result is a bang-bang characterization:

- If N is vertically ordered, the unique equilibrium has $a = 0$ for all types, so $L = 0$.
- If N is not vertically ordered, for any $\theta, \beta \in (0, 1)$ there are primitives and type shares such that $\Pr(a = 1) > \beta$ in some equilibrium. When θ is close to 1, loss can approach 1.

Interpretation. Nested controls shut down confounding recursively. Horizontally differentiated controls can create confounding patterns that fool each other.

3.5 The horizontal-differentiation mechanism

Example 3.2 has two controls x_1, x_2 and two data types:

$$C_1 = \{1\}, \quad C_2 = \{2\}.$$

Each type conditions his action on a different variable. A candidate equilibrium is

$$a = x_i \quad \text{for type } i.$$

Because the aggregate action a is a common consequence of x_1 and x_2 , conditioning on a can create correlations between otherwise unrelated variables. Each type's behaviour becomes a confounder for the other type.

Interpretation. This is the main reason more controls are not always better in equilibrium. Adding a horizontally different control can create new data patterns that reinforce false causal beliefs.

3.6 Heterogeneous preferences: $\gamma \in (0, 1)$

When $p(t = 1) = \gamma$, the rational benchmark is $a = t$. The analysis focuses especially on the restricted domain

$$y \perp x \mid t.$$

This means t is sufficient for the outcome; x is only a proxy for t .

For vertically ordered types, the paper shows

$$L \leq \gamma(1 - \gamma),$$

and this upper bound is tight.

For non-vertically ordered types, the restricted-domain upper bound becomes

$$L \leq \max\{\gamma, 1 - \gamma\},$$

again tight under the stated conditions. Without the restricted-domain assumption, full error can be sustained:

$$\Pr(a \neq t) = 1$$

for suitable primitives.

Interpretation. Preference heterogeneity creates a genuine reason for behaviour to vary with t . Equilibrium no longer necessarily drives loss to zero, but vertical ordering still sharply limits the worst case.

3.7 Adjustment without conditioning

Section 5 distinguishes:

- C_i : variables whose realizations the type observes at the moment of choice;
- D_i : variables for which the type has long-run statistical data and can therefore adjust.

The assumption is $C_i \subseteq D_i$. The perceived causal effect becomes

$$\Delta_i(x) = \sum_{x_{D_i \setminus C_i}} p(x_{D_i \setminus C_i} \mid x_{C_i}) [E_p(y \mid a = 1, x_{D_i}) - E_p(y \mid a = 0, x_{D_i})].$$

Interpretation. A decision-maker can statistically adjust for a variable even if he does not observe its current realization. This is closer to some real data-use problems, where historical statistics are available but current states are not.

3.8 Vertical ordering in the adjustment model

Define a binary relation P over data types:

$$iPj \iff D_i \supseteq C_j.$$

The set of data types is vertically ordered if P is complete and quasitransitive.

Interpretation. Type i is above type j if i controls for every variable on which j conditions. This is the right analogue of nested control sets when controls and conditioning variables differ.

3.9 Consequential actions

Section 6 allows a to have a true separable causal effect on y . Introduce an unobserved variable z and suppose

$$E_p(y \mid a, z) = \beta a + (1 - \beta)z,$$

where $\beta \in (0, 1)$ is the true causal effect of a . The expected welfare loss becomes

$$\gamma p(a = 0 \mid t = 1)(\theta + \beta) + (1 - \gamma)p(a = 1 \mid t = 0)(\theta - \beta).$$

The upper bounds from the previous sections are multiplied by $1 - \beta$.

Interpretation. A real causal benefit reduces the welfare cost of mistaken inference because part of the action-outcome correlation is actually causal. But the equilibrium logic remains the same.

4 Identification and theoretical design

4.1 What the paper identifies

The paper is not an empirical identification exercise. It identifies worst-case equilibrium bounds on welfare loss generated by bad controls in an endogenous database.

- The primitives are distributions over preference types and exogenous variables.
- Data types differ in the control variables they use.
- Strategies determine the public action distribution.
- The public distribution determines perceived causal effects.
- Equilibrium requires strategies to be best replies to those perceived effects.

Interpretation. The key theoretical object is a fixed point between behaviour and causal beliefs.

4.2 Why the bounds are useful

- They show when equilibrium discipline greatly reduces the welfare cost of bad causal inference.
- They show when this discipline fails.
- They separate the role of preference heterogeneity from the role of control-set geometry.
- They explain why nested data access is safer than horizontally fragmented expertise.

Interpretation. The paper's policy-relevant message is qualitative rather than numerical: the organization of data access and expertise matters for the aggregate harm from bad causal reasoning.

4.3 Summary of main bounds

Environment	Control geometry	Worst-case implication
Homogeneous preferences, $\gamma = 0$	Vertically ordered	Unique equilibrium has no error; loss = 0
Homogeneous preferences, $\gamma = 0$	Not vertically ordered	Loss can approach non-equilibrium worst case
Heterogeneous preferences, $y \perp x \mid t$	Vertically ordered	Loss $\leq \gamma(1 - \gamma)$; tight
Heterogeneous preferences, $y \perp x \mid t$	Not vertically ordered	Loss $\leq \max\{\gamma, 1 - \gamma\}$; tight under conditions
Unrestricted heterogeneous preferences	Not vertically ordered	Full error, $\Pr(a \neq t) = 1$, can be sustained
Adjustment model	Vertically ordered under P	Same bang-bang logic as baseline
Consequential action, true effect β	Same as above	Bounds are multiplied by $1 - \beta$

4.4 What the theory does not fully settle

- It does not provide a fully dynamic learning foundation for all equilibria, although the equilibrium concept is motivated by a steady-state learning story.
- It focuses on binary actions and preference types.
- The distribution over data types is independent of t and x .
- Some tightness results require constructed distributions rather than natural empirical parameterizations.
- Extensions with correlated data types, arbitrary vertical spaces under heterogeneous preferences, and richer dynamic learning are left open.

Interpretation. The paper is a worst-case equilibrium theory. Its purpose is to clarify mechanisms and bounds, not to estimate a particular empirical welfare loss.

5 Tables and figures

The paper contains no empirical tables or plotted figures. Following the format used in the earlier theoretical notes, I directly extract and group the key original equation, definition, example, and proposition panels from the paper.

5.1 Model primitives and equilibrium

Read:

- Panel (a) shows that the public action distribution is generated by the weighted strategies of all data types.
- The perceived causal effect $\Delta_i(x)$ is the difference in conditional expected outcomes after controlling for C_i .
- Panel (b) computes perceived causal effects in the one-control example and gives the equilibrium best-response condition.
- The example makes clear why $\Delta_i(x)$ depends on other types' strategies.

Economic message. The main novelty is endogeneity of the data. The observed conditional correlations are not fixed objects; they are generated by behavioural responses to previous correlations.

the conditional-independence property $y \perp a \mid (t, x)$, and hence factorizes as $p(t, x, a, y) = p(t, x)p(a \mid t, x)p(y \mid t, x)$, where $p(t, x)$ and $p(y \mid t, x)$ are exogenous, while $p(a \mid t, x)$ is endogenous, representing the DM's average behaviour across data types:

$$p(a \mid t, x) = \sum_{i \in N} \lambda_i \sigma_{t,i}(a \mid x_{C_i})$$

Thus, the public data that a DM of any type relies on is *aggregate*, representing the endogenous choices of *all* types. In what follows, I take it for granted that σ is implicit in p , without notating this explicitly.

Since a DM of data type i believes that C_i is a valid set of controls, he regards $p(y \mid a, x_{C_i})$ as a proper estimate of the probabilistic consequence of choosing a , given his observation of x_{C_i} . His perceived causal effect of a on y given x is

$$\Delta_i(x) = E_p(y \mid a = 1, x_{C_i}) - E_p(y \mid a = 0, x_{C_i}) \quad (1)$$

If the DM had long-run data about all exogenous variables (including t), he could control for all of them, and thus correctly infer the action's null causal effect. In contrast, our DM may end up

(a) Model and perceived causal effect

types play $a = 1$ whenever $t = 1$, $p(t = 1 \mid a = 0, x) = 0$ for every x . It follows that

$$\begin{aligned} \Delta_1(1) &= p(t = 1 \mid a = 1, x = 1) = \frac{\gamma q}{\gamma q + (1 - \gamma)(1 - q)[\lambda_1 a_1(1) + \lambda_2 a_2]} \\ \Delta_1(0) &= p(t = 1 \mid a = 1, x = 0) = \frac{\gamma(1 - q)}{\gamma(1 - q) + (1 - \gamma)q[\lambda_1 a_1(0) + \lambda_2 a_2]} \\ \Delta_2 &= p(t = 1 \mid a = 1) = \frac{\gamma}{\gamma + (1 - \gamma)[\lambda_1(q a_1(0) + (1 - q)a_1(1)) + \lambda_2 a_2]} \end{aligned}$$

Note that by definition, Δ_2 is a weighted average of $\Delta_1(1)$ and $\Delta_1(0)$. Therefore, if type 2 weakly prefers to play $a \neq t$, so must type 1 with some probability. We will revisit these formulas in Example 4.2. $\square$

As the example demonstrates, the DM's estimate of the causal effect of a can be sensitive to the strategy profile σ . This observation suggests that to define subjective optimization with respect to the causal belief one extracts from an endogenous database, an equilibrium approach is called for.

Definition 1 (Equilibrium). Let $\varepsilon \in (0, \frac{1}{4})$. A strategy profile $\sigma = (\sigma_1, \dots, \sigma_n)$ is an ε -equilibrium if for every $i = 1, \dots, n$ and every $t, x, a', \sigma_{t,i}(a' \mid x) > \varepsilon$ only if

$$a' \in \arg \max_a \{E_p(y \mid a, x_{C_i}) - \theta \cdot \mathbb{I}[a \neq t]\}$$

(b) Example 2.1 and equilibrium definition

5.2 Welfare loss and vertical ordering

Read:

- Panel (a) gives the expected welfare loss formula: the taste cost θ multiplied by the probability of choosing $a \neq t$.
- Panel (b) defines vertical ordering and states Proposition 1.
- With homogeneous preferences and vertically ordered control sets, equilibrium forces fully eliminate the welfare cost of bad controls.

Economic message. This is the first major protection result. Nested controls make the most sophisticated type's behaviour correct, then recursively remove the confounding faced by less sophisticated types.

Definition 3 (Vertically ordered types). The set of data types N is vertically ordered if types can be enumerated such that $C_1 \supset \dots \supset C_n$.

When N is vertically ordered, lower-indexed data types control for a larger set of variables. In particular, type i controls for every variable that type $j > i$ conditions on. In this case, data types are naturally ranked in terms of how close they are to the ideal of controlling for all potential confounders (since the DM never controls for t , no type attains this ideal).

As mentioned in the Introduction, vertical ordering fits situations in which all DM types agree on what they consider to be relevant controls; however, they have limited degrees of data access. Variables are ordered in terms of how easy it is to get access to data about them; DM types can be viewed as rungs on this ladder. More broadly, vertical ordering of DM types resonates with other areas of economic theory that classify agents according to some linear ordering (ranking preference types by a willingness-to-pay scalar, ranking information types by the quality of their signal, etc.). Such typologies are attractive to economic theorists because they are interpretable and generate sharp results. The same holds in this paper. DMs with larger sets of control variables are intuitively closer to the ideal of correct causal inference. Therefore, they are intuitively more "sophisticated" (though we will have opportunities to be reminded that adding controls is not necessarily beneficial). And as our analysis will now begin to show, vertically ordered DM spaces generate strong results about the equilibrium decision costs of bad controls.

Proposition 1. Let $\gamma = 0$. Suppose N is vertically ordered. Then, the unique equilibrium is for all DM types to play $a = 0$ with probability one. In particular, the DM's expected welfare loss is zero.

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Definition 2 (Expected welfare loss). Given a strategy profile σ , the DM's expected welfare loss is

$$\theta \sum_{t,x} p(t, x) \sum_{i \in N} \lambda_i \sigma_{t,i}(a \neq t \mid x) \quad (2)$$

(a) Expected welfare loss

(b) Vertical ordering and Proposition 1

5.3 Horizontally differentiated controls

Read:

- Panel (a) shows the condition in Example 3.2 under which two horizontally differentiated data types sustain erroneous behaviour.

- Proposition 2 generalizes the example: without vertical ordering, equilibrium loss can approach the non-equilibrium benchmark.
- Panel (b) shows the chess-and-math heterogeneous-preferences example, where equilibrium erodes the perceived causal effect.

Economic message. Horizontally differentiated types can fool each other. Each type’s action varies with one variable and creates a residual correlation that the other type misreads as causal.

In particular, when the factors x_1 and x_2 are independently and uniformly distributed, the equilibrium condition simplifies into

$$\delta_{11} - \delta_{10} \geq \frac{3}{2}\theta \geq \delta_{01} - \delta_{00}$$

and the welfare loss is $\frac{1}{2}\theta \leq \frac{1}{3}$. A more extreme specification is $p_{11} \approx 1$ and $\delta_{i,x_2} = x_1 x_2$. In this case, the equilibrium condition holds for every $\theta < 1$, and the equilibrium expected welfare loss is approximately θ , which itself can be arbitrarily close to the non-equilibrium benchmark of 1.

The intuition behind this result is that since type i varies its action with x_i yet fails to control for x_j , each type creates a confounding effect that “fools” the other type. Type i is vulnerable to interpreting the residual correlation between a and y after controlling for x_i —which exists because of type j ’s strategy—as a causal effect. While the correlation between x_1 and x_2 can exacerbate the DM’s welfare loss, it is not necessary for it: As we saw, the loss can arise even when the two factors are independent. The reason is that although the two data types condition their actions on independent exogenous variables, their subjective causal estimates involve conditioning on a —a variable whose distribution records firms’ aggregate behaviour. Since this variable is a common consequence of x_1 and x_2 , conditioning on it creates a correlation between otherwise-independent variables.

The equilibrium welfare loss is non-monotone with respect to the data types’ sets of control variables. For example, suppose $C_1 = \{1\}$ and $C_2 = \emptyset$ —i.e. type 2 now does not control for any variable. In this case, the type space is vertically ordered; and by Proposition 1, neither data type will commit an error in equilibrium. It follows that expanding one type’s set of control variables can be detrimental for all types’ welfare. $\square$

The following result generalizes this example.

Proposition 2. Let $\gamma = 0$. Suppose N is not vertically ordered. Then, for any $\theta, \beta \in (0, 1)$, there exist λ and $(p(x, y))$ such that $\Pr(a = 1) > \beta$ in some equilibrium. In particular, when $\theta \approx 1$, the equilibrium welfare loss can be arbitrarily close to 1.

(a) Horizontal controls and Proposition 2

I establish the uniqueness of equilibrium in this setting, and characterize the DM’s equilibrium welfare loss. The DM’s estimated causal effect of a on y is

$$\Delta = p(y = 1 | a = 1) - p(y = 1 | a = 0)$$

Since the DM’s intrinsic payoff from playing $a = 1$ increases with t , we must have $a_1 \geq a_0$ in equilibrium. Now obtain explicit expressions for the terms that define Δ :

$$\begin{aligned} p(y = 1 | a = 1) &= \frac{\gamma \cdot a_1 \cdot \delta_1 + (1 - \gamma) \cdot a_0 \cdot \delta_0}{\gamma \cdot a_1 + (1 - \gamma) \cdot a_0} \\ p(y = 1 | a = 0) &= \frac{\gamma \cdot (1 - a_1) \cdot \delta_1 + (1 - \gamma) \cdot (1 - a_0) \cdot \delta_0}{\gamma \cdot (1 - a_1) + (1 - \gamma) \cdot (1 - a_0)} \end{aligned}$$

A simple calculation establishes that since $\delta_1 > \delta_0$ and $a_1 \geq a_0$, we must have $\Delta \geq 0$. This in turn implies that $a_1 \geq 1 - \varepsilon$ in ε -equilibrium, because when $t = 1$, the DM perceives no conflict between his intrinsic taste for playing $a = t$ and the estimated effect of his choice on y . Plugging the known expressions for a_1 and δ_1 and taking the $\varepsilon \rightarrow 0$ limit, we obtain

$$\Delta = \frac{\gamma}{\gamma + (1 - \gamma) \cdot a_0}$$

If $a_0 \leq \varepsilon$ in ε -equilibrium, then $\Delta \rightarrow 1$ in the $\varepsilon \rightarrow 0$ limit. But then $\Delta > \theta$, hence playing $a = 1$ at $t = 0$ is the unique subjective best-reply. Therefore, the ε -equilibrium requirement is that $a_0 > \varepsilon$, a contradiction. It follows that $a_0 > 0$ in equilibrium. There are two cases to consider.

Case 1: $a_0 \in (0, 1)$. This requires the DM to be indifferent between the two actions—i.e. $\Delta = \theta$. Therefore, $\gamma < \theta$ and

$$a_0 = \frac{\gamma(1 - \theta)}{(1 - \gamma)\theta}$$

Since the DM only commits an error in equilibrium when $t = 0$, his expected equilibrium welfare loss is

$$\theta \cdot (1 - \gamma) \cdot a_0 = \gamma(1 - \theta) < \gamma(1 - \gamma)$$

(b) Chess-and-math equilibrium calculations

5.4 Heterogeneous-preference bounds

Read:

- Lemma 1 says that with vertical ordering and $y \perp x | t$, every type’s estimated effect has the same sign in equilibrium.
- Proposition 3 gives the tight bound $\gamma(1 - \gamma)$.
- Proposition 4 pins down the one-control, two-type case and shows the loss decreases with the share of sophisticated types.
- Proposition 5 shows that without vertical ordering, the restricted-domain upper bound rises to $\max\{\gamma, 1 - \gamma\}$.

Economic message. With preference heterogeneity, equilibrium protection is no longer complete, but nested controls still sharply limit the worst case.

restricted by $y \perp x \mid t$.

Lemma 1. Suppose N is vertically ordered. If $y \perp x \mid t$, then in equilibrium, $a_t = 1$ and $\Delta_t(x) \geq 0$ for every data type i .

Thus, when data types are vertically ordered, the DM's estimated causal effect of a on y is always non-negative in equilibrium, regardless of his data type and the realization of x . While data types may disagree on the magnitude of the causal effect of a on y , they all agree on its sign. Consequently, the DM must always play $a = 1$ when $t = 1$ in any equilibrium.

As with Proposition 1, the proof of Lemma 1 proceeds by induction on the set of data types, starting with type 1, whose set of controls is the largest. Although this type controls for every x variable the other data types condition on, this does not mean he is immune to neglecting confounders, because he cannot control for t . Furthermore, since this type varies his behaviour with t , he exerts a "confounding externality" (of the kind we encountered in Example 3.2) on the other data types. This makes the inductive proof more intricate.

Proposition 3. Suppose N is vertically ordered. If $y \perp x \mid t$, then the DM's expected welfare loss in equilibrium is at most $\gamma(1 - \gamma)$.

Example 4.1 established the tightness of this upper bound. Proposition 3 also means that across all distributions that satisfy $y \perp x \mid t$, the expected welfare loss is at most $\frac{1}{4}$ —compared

(a) Lemma 1 and Proposition 3

The following result provides a more general equilibrium characterization when there is a single potential control variable. In this case, the data types are vertically ordered because there the only possible types are $\{1\}$ and $\emptyset$. The result is easily extendible to any setting with $K > 1$, $n = 2$, and $C_2 = \emptyset$. I impose minor restrictions on the primitives (the result does not rely on them; they merely shorten the proof).

Proposition 4. Let $K = 1$ and $n = 2$, such that $C_1 = \{1\}$ and $C_2 = \emptyset$. Suppose that $p(t, x)$ has full support and that $\theta > \gamma$. Then, the equilibrium welfare loss is uniquely determined for any given $p(t, x)$, λ , θ . It is weakly below $\gamma(1 - \theta)$, and weakly decreasing in λ_1 .

This result establishes that when there is a single potential control variable, the equilibrium welfare loss is pinned down by the primitives. It is always below the upper bound given by Proposition 3, and it decreases with the fraction of sophisticates. As we saw in Section 3, the latter property does not hold in general. Note that Proposition 4 does not rule out multiple equilibria. Specifically, when the likelihood ratio $p(x \mid t = 0)/p(x \mid t = 1)$ is sufficiently close to one for every x , there is a continuum of equilibria, in all of which the welfare loss is exactly $\gamma(1 - \theta)$. (In any equilibrium, the "sophisticated" type 1 must make errors with positive probability.)

When data types are not vertically ordered, the tight upper bound on the DM's expected welfare loss (under the restriction $y \perp x \mid t$) is significantly higher.

Proposition 5. Suppose N is not vertically ordered. If $y \perp x \mid t$, then the DM's expected welfare loss in equilibrium is at most $\max(\gamma, 1 - \gamma)$. When $|X_k| \geq 3$ for all k , this upper bound can be approximated arbitrarily well, by appropriately selecting θ , λ and $(p(x, y \mid t))$.

(b) Propositions 4 and 5

5.5 Unordered types and adjustment without conditioning

Read:

- Proposition 6 shows that if types are not vertically ordered and the distribution is unrestricted, full error can be sustained.
- Panel (b) shows the adjustment formula: the DM conditions on C_i but averages over variables in $D_i \setminus C_i$.
- Example 5.1 illustrates a bad pre-treatment control: adjusting for a common consequence of two variables can create a biased estimate.

Economic message. The distinction between conditioning and adjustment matters, but the deeper problem is still the same: an apparently reasonable control can distort the causal estimate.

The DM's estimated causal effect of switching from $a = 0$ to $a = 1$ (given x) is

$$\Delta_t(x) = \sum_{x_D \in C} p(x_{D \setminus C} \mid x_C) [E_p(y \mid a = 1, x_D) - E_p(y \mid a = 0, x_D)] \quad (4)$$

Thus, controlling for x_D involves conditioning on x_C and adjusting for $x_{D \setminus C}$.

Comment: Subjective state spaces. The perceived causal effect given by (4)—and by implication, the simpler formula (1)—can be interpreted traditionally in terms of the Savage framework, where the state space itself is *subjective*. According to this interpretation, X_D is type t 's subjective state space and X_C is his set of signals. The novelty here is that while the state space is subjective, the DM's belief is a projection of the *objective* distribution p on his subjective state space. Moreover, unlike the standard Savage model, the stochastic mapping from the DM's subjective states to outcomes is affected by the behaviour of other DM types, hence it is an *endogenous* object. In my opinion, these deviations from the Savage framework are so drastic that they justify my decision to avoid the Savage terminology altogether in the paper's formal exposition.

Example 5.1 (Adjusting for an irrelevant variable). This example illustrates the danger of excessive controlling for "pre-treatment" variables, independently of equilibrium considerations. It is adapted from Cinelli *et al.* (2024), a guide to "good and bad controls" that, following Pearl (2009), makes use of the formalism of directed acyclic graphs (DAGs). Let $t = 0$ with certainty, and suppose that the true causal structure underlying p is given by the DAG

$$a \leftarrow x_1 \rightarrow x_3 \leftarrow x_2 \rightarrow y$$

All variables take values in $\{0, 1\}$; x_1 and x_2 are uniformly distributed; $y = x_2$ and $x_3 = x_1 x_2$ with certainty; and $p(a = x_1 \mid x_1) = 1 - \varepsilon$ for all x_1 , where $\varepsilon \approx 0$. The objective causal effect of a on y is null because the DAG includes no causal path from a to y . Therefore, $E_p(y \mid a = 1) - E_p(y \mid a = 0)$ is a correct formula for the null objective causal effect. In other words, there is no need to control for any of the x variables.

Suppose, however, that one of the DM types has $C = \emptyset$ and $D = \{3\}$ —i.e. he does not condition on any variable, while adjusting for x_3 .⁵ The type's estimated causal effect is

$$\sum_{x_3} p(x_3) [E_p(y \mid a = 1, x_3) - E_p(y \mid a = 0, x_3)] \quad (5)$$

(a) Proposition 6

(b) Adjustment and a bad-control example

$(p(x, y \mid t))$. It shows that, in this case, the gap between equilibrium and non-equilibrium upper bounds on the DM's welfare loss disappears.

Proposition 6. Suppose N is not vertically ordered. For every $\gamma, \theta \in (0, 1)$, there exist λ and $(p(x, y \mid t))$ for which there is an equilibrium in which $\Pr(a \neq t) = 1$.

The results in this section leave three open problems. First, does the characterization in Proposition 4 extend to arbitrary vertically ordered type spaces? Second, does the upper bound $\gamma(1 - \gamma)$ obtained for vertically ordered types extend to distributions n for which $y \perp x \mid t$?

5.6 Extended vertical ordering and consequential actions

Read:

- Definition 4 adapts equilibrium to the adjustment model.
- Definition 5 replaces simple set inclusion by the relation P : type i is above type j if D_i contains every variable that j conditions on.
- Propositions 7 and 8 reproduce the bang-bang result in the extended model.
- The consequential-action extension adds a true causal effect β and scales the welfare bounds by $1 - \beta$.

Economic message. The vertical-ordering idea is robust. It is not merely an artifact of the assumption that a type can condition on every variable he controls for.

Definition 4. A strategy profile $\sigma = (\sigma_1, \dots, \sigma_n)$ is an ε -equilibrium if for every $i = 1, \dots, n$ and every $t, x, a', \sigma_{-i}(a' | x) > \varepsilon$ only if

$$a' \in \arg \max_a \left\{ \sum_{x_{D_i \setminus C_i}} p(x_{D_i \setminus C_i} | x_{C_i}) E_p(y | a, x_{C_i}) - \theta \cdot \mathbf{1}[a \neq t] \right\}$$

I now extend the notion of vertically ordered types. Define a binary relation P over data types: iPj if $D_i \supseteq C_j$. The meaning of iPj is that data type i controls for every variable that type j conditions on. Since $D_i \supseteq C_i$ for every $i \in N$, P is reflexive. Let P^* be the asymmetric (strict) part of P —i.e. iP^*j if iPj and not jPi . Following Sen (1969), P is *quasitransitive* if P^* is transitive.

Definition 5. The set N is vertically ordered if the binary relation P is complete and quasitransitive.

When $C_i = D_i$ for every $i \in N$, this definition collapses to Definition 3.

The following observation is standard. We say that type i is P^* -undominated in a set of types M , if there is no $j \in M$ such that jP^*i .

Remark 1. Suppose P is complete and quasitransitive. Then, N can be partitioned into L classes, $N_1, \dots, N_L$, such that: (i) N_1 consists of all P^* -undominated types in N ; and (ii) for every $\ell > 1$, N_ℓ consists of all P^* -undominated types in $N \setminus (\cup_{h < \ell} N_h)$.

The partition induced by a complete and quasitransitive P is the extended model's analogue of vertical ordering of types.

The following results extend the worst-case analysis of Section 3 (homogenous preferences).

Proposition 7. Let $\gamma = 0$. Suppose N is vertically ordered. Then, the unique equilibrium is for all DM types to play $a = 0$ with probability one. In particular, the DM's expected welfare loss is zero.

Proposition 8. Let $\gamma = 0$. Suppose N is not vertically ordered. Then, for any $\theta, \beta \in (0, 1)$, there exist λ and $(p(x, y))$ such that $\Pr(a = 1) > \beta$ in some equilibrium. In particular, when $\theta \approx 1$, the equilibrium welfare loss can be arbitrarily close to 1.

(a) Extended vertical ordering

6. CONSEQUENTIAL ACTIONS

So far, I have focused on the extreme case in which the DM's action has no objective causal effect on the outcome. This facilitated the definition of the DM's equilibrium welfare loss due to poor controls, relative to the rational-expectations benchmark. This section extends the analysis to situations in which actions have an *additively separable* causal effect on outcomes. This kind of separability is commonly assumed in empirical research. Therefore, it is sensible to assume it in our context as well.

Define an unobservable variable z that takes values in $[0, 1]$. This variable is a consequence of (t, x) , *independently* of a , just as y was in the basic model. The outcome y is purely caused by a and z (i.e. $y \perp (t, x) | (a, z)$), such that

$$E_p(y | a, z) = \beta a + (1 - \beta)z \quad (6)$$

where $\beta \in (0, 1)$ quantifies the true causal effect of a on y . The DM is unaware of the relation (6), nor does he know β . As before, he forms beliefs by examining the observed joint distribution of a, y , and his set of control x variables. Nevertheless, we can express type i 's estimated causal effect of switching from $a = 0$ to $a = 1$ on y given x as $\beta + (1 - \beta)\Delta_i^z(x)$, where

$$\Delta_i^z(x) = \sum_{x_{D_i \setminus C_i}} p(x_{D_i \setminus C_i} | x_{C_i}) [E_p(z | a = 1, x_{D_i}) - E_p(z | a = 0, x_{D_i})]$$

This expression makes use of the extended notion of DM types presented in Section 5, which allows D_i to be a strict superset of C_i . Since $z \perp a | (t, x)$, the equilibrium analysis of $\Delta_i^z(x)$ and how it relates to the DM's strategy is the same as the analysis of $\Delta_i(x)$ in the previous sections.

It follows that the only thing that needs adjustment is the definition of the DM's welfare loss. The optimal rational-expectations action maximizes $\beta a - \theta \cdot \mathbf{1}[a \neq t]$, because a has no causal effect on z , such that the only effect of a on y is via the direct channel parameterized by β . Therefore, the expected welfare loss given a joint distribution p is

$$\gamma \cdot p(a = 0 | t = 1) \cdot (\theta + \beta) + (1 - \gamma) \cdot p(a = 1 | t = 0) \cdot (\theta - \beta) \quad (7)$$

(b) Consequential actions

5.7 Extreme sports example

Read:

- The example studies a risky lifestyle choice with a genuine causal health effect and age as a confounder.
- Sophisticated types control for age and correctly infer the health benefit.
- Naive types do not control for age and can be led by endogenous correlations to follow their tastes.
- A larger share of sophisticated types can reduce naive errors through an equilibrium externality.

Economic message. Educating some people to reason better about causality can have a multiplier effect because their choices alter the correlations that naive decision-makers learn from.

Suppose that $a = 1$ means that the DM refrains from a potentially pleasurable, yet dangerous activity. For the sake of the example, this activity is extreme sports. The outcome $y = 1$ represents good health—specifically, lack of injuries. Let x represent the DM's age ($x = 0$ indicates a young DM). Let t represent the DM's intrinsic taste for the activity: $t = 0$ means that the DM likes extreme sports. Let $\theta < \frac{1}{2}$.

The objective distribution p is described as follows. First, $p(x = 1) = \frac{1}{2}$. Second, $p(t = x | x) = q$ for all x , where $q \in (\frac{1}{2}, 1)$. That is, young (old) people tend to like (dislike) extreme sports. Finally, let $p(y = 1 | a, x) = \frac{1}{2}(a + 1 - x)$. That is, old age and extreme sports increase the propensity for injuries. The joint distribution p is consistent with the DAG



Data type 1 controls for x . This type correctly estimates the causal health effect of switching from $a = 0$ to $a = 1$ to be $\frac{1}{2}$. Since $\theta < \frac{1}{2}$, this DM data type will rationally play $a = 1$, independently of t and x . Data type 2 does not control for x .⁷ This DM chooses a to maximize

$$p(y = 1 | a) - \theta \cdot \mathbf{I}[a \neq t] = \frac{1}{2}[a + 1 - p(x = 1 | a)] - \theta \cdot \mathbf{I}[a \neq t]$$

Let us analyse equilibria in this example.

Claim 2. The rational-choice benchmark can be sustained in equilibrium.

Proof: To prove this claim, recall that data type 1's strategy is $\sigma_1(a = 1 | t, x) = 1$ for all t, x . Denote $\sigma_2(a = 1 | t) = a_t$. Then,

$$\begin{aligned} p(x = 1 | a = 1) &= \frac{\lambda_1 + \lambda_2[q a_1 + (1 - q)a_0]}{2\lambda_1 + \lambda_2[a_1 + a_0]} \\ p(x = 1 | a = 0) &= \frac{1 - q a_1 - (1 - q)a_0}{2 - a_1 - a_0} \end{aligned}$$

First, let us guess

$$p(x = 1 | a = 1) - p(x = 1 | a = 0) < \frac{1}{2} - \theta$$

Then, $a = 1$ is optimal for data type 2 regardless of t . In this case, we need to consider perturbed strategies to ensure that $p(x = 1 | a = 0)$ is well-defined. Since a_0 and a_1 are arbitrarily close to 1, we obtain $p(x = 1 | a = 1) \approx \frac{1}{2}$. We can also set the perturbations such that $p(x = 1 | a = 0) = \frac{1}{2}$. It follows that it is always possible to sustain the guess in equilibrium, such that the DM will commit no error. ■

However, equilibria that exhibit decision errors are also possible.

Claim 3. Assume

$$\theta > \frac{1}{2} - \frac{2q - 1}{1 + \lambda_1} \quad (8)$$

(a) Claims 2 and 3

Then, there is an equilibrium in which type 2 always plays $a = t$.

Proof: To verify this claim, let us guess

$$p(x = 1 | a = 1) - p(x = 1 | a = 0) > \frac{1}{2} - \theta$$

Then, data type 2 will play $a_t \equiv t$ in equilibrium. Plugging this into the expressions for $p(x = 1 | a)$, we obtain

$$p(x = 1 | a = 1) - p(x = 1 | a = 0) = \frac{\lambda_1 + \lambda_2 q}{2\lambda_1 + \lambda_2} - (1 - q)$$

Condition (8) means that this expression exceeds $\frac{1}{2} - \theta$, thus confirming that the guess is consistent with equilibrium. ■

What sustains this equilibrium is the positive correlation between age and preferences. Young DMs like extreme sports more than old DMs, and since the DM chooses according to his intrinsic taste with probability $\lambda_2 > 0$, there is positive correlation between doing extreme sports and young age. In turn, this softens the negative correlation between a and y , to an extent that makes it subjectively optimal for type-2 DMs to follow their taste.

The expected welfare loss in this equilibrium is

$$\frac{1}{2} \cdot \lambda_2 \cdot \left(\frac{1}{2} - \theta \right) < \left(q - \frac{1}{2} \right) \frac{\lambda_2}{2 - \lambda_2}$$

The R.H.S of this inequality represents the maximal welfare loss in this setting. It increases with the fraction of type 2. There are two forces behind this observation. First, higher λ_2 obviously means that there are more DMs in the population who are prone to error. Second, type-1 DMs

(b) Welfare loss and equilibrium externality

6 Short summary

- The paper models decision-makers who infer causal effects from observational correlations using controls.
- The dataset is endogenous because previous decision-makers' actions help generate the correlations observed by future decision-makers.
- Each data type uses a different set of controls.
- The decision-maker treats his controlled correlation as a causal effect and best-responds to it.
- Equilibrium is a steady state in which action frequencies and perceived causal effects are mutually consistent.
- The welfare loss is the taste cost from choosing an action that does not match the true preference type.
- If preferences are homogeneous and control sets are vertically ordered, the unique equilibrium has no welfare loss.
- If control sets are not vertically ordered, horizontally differentiated types can create confounding patterns that fool each other.
- With preference heterogeneity and $y \perp x | t$, vertical ordering gives the tight upper bound $\gamma(1 - \gamma)$.
- Without vertical ordering, the restricted-domain bound becomes $\max\{\gamma, 1 - \gamma\}$, and full error is possible without the restriction.
- The model extends to cases where decision-makers adjust for variables they do not observe at the moment of choice.

- The model also extends to cases where the action has a true separable causal effect; the welfare bounds are scaled by $1 - \beta$.
- The main message is that bad controls may be less harmful in equilibrium than in a fixed-data benchmark, but only when control sets are nested.

7 Conclusion

7.1 Core conclusion

Spiegler develops an equilibrium theory of behavioural causal inference. Decision-makers use available controls to interpret correlations as causal effects, but their actions also shape the correlations from which later decision-makers learn. This feedback can either protect decision-makers from bad controls or amplify their mistakes.

7.2 Economic intuition

When everyone uses nested controls, the most informed types stop creating misleading variation, and this stabilizes the inference problem for less informed types. When types use horizontally different controls, each type's behaviour can create confounding for another type. In that case, equilibrium does not necessarily discipline bad causal inference.

7.3 Takeaway

The cost of bad controls depends not only on individual statistical mistakes but also on the organization of data access and expertise in the population. Nested data access is relatively safe. Fragmented, horizontally differentiated data access can make flawed causal beliefs self-sustaining.